

Victor Nico DJIEMBOU TIENTCHEU

Senior Data Scientist | Machine Learning Engineer

Yaoundé, Cameroon — Open to remote roles worldwide (CET / GMT+1 overlap)

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Summary

Data Scientist and full-stack engineer with an MSc in Data Science and 6+ years of applied experience in machine learning, NLP and production software systems. Specialised in predictive modelling, graph-based feature engineering, retrieval-augmented generation (RAG) and explainable ML. Comfortable owning a problem end-to-end: data, modelling, deployment and database administration. Bilingual French-English. Winner of the CONIA 2025 Hackathon; recipient of the Best Poster and Best Project Awards at I3AFD 2026.

Core Skills

Machine Learning & Data Science: Python, scikit-learn, XGBoost, PyTorch, TensorFlow, NumPy, Pandas, NetworkX; classification, regression, clustering, graph ML, cost-sensitive learning, feature engineering, SHAP, LIME.

NLP & LLM / RAG: LlamaIndex, LangChain, FAISS, Hugging Face Transformers, BAAI/bge-m3 embeddings, cross-encoder reranking, Mistral, BERT/Roberta fine-tuning, prompt engineering, evaluation pipelines.

Databases & Data Engineering: PostgreSQL, MongoDB, Neo4j, Supabase; advanced SQL, CTEs, window functions, query tuning, schema migration.

Software & MLOps: Vue.js / Nuxt 3, Node.js, Express, Flask, FastAPI, Docker, Git, CI/CD (GitHub Actions), PM2, Nginx, REST APIs, VPS deployment.

Languages: French (native), English (professional working proficiency).

Professional Experience

Technical Project Lead

2025 – Present

DBA Solutions SAS, Argenteuil, France / Douala, Cameroon (remote)

- Lead a technical project within a French-Cameroonian IT services company delivering software engineering, database administration and data solutions to SMEs and public-sector clients.
- Drive high-quality software delivery: performance tuning, high-availability setups, backup strategies, security and production incident response.
- Lead the design and development of a high-availability Management Information System (MIS) on a meta-data-driven architecture to improve scalability and maintainability.

Lead Trainer

2023 – Present

Independent, Yaoundé, Cameroon (remote)

- Design and deliver technical training programmes in data science, machine learning, full-stack web development and Linux systems programming to university students and working professionals.
- Produce reusable teaching material (courses, handouts, interactive notebooks) in French and English.
- Mentor students on applied research, competition projects and open-source contributions.

Lead Software Developer & Data Analyst

2021 – 2026

TAKENCO, Yaoundé, Cameroon

- Designed and deployed the company's centralised data architecture (geo-positions, deliveries, routing, traffic, timing) for an urban mobility and last-mile delivery platform.
- Built interactive dashboards to expose operational KPIs and support management decisions.
- Implemented versioned REST APIs to serve data to product and operations teams.
- Improved team coordination and development velocity; stack: Vue.js, Node.js, Express, Flask, PostgreSQL, MongoDB, Docker.

Full-Stack Developer (Freelance / Contract)

2018 – 2023

Independent, Remote — Cameroon / Belgium / France / USA

- Delivered 10+ production web applications for SME clients, including data pipelines and automated reporting.
- Designed robust backend architectures (REST APIs, event-driven components, CI/CD).
- Applied principles of high availability, scalability, data-driven and domain-driven design, monitoring and alerting, SEO and accessibility.

Selected Projects & Research

EvalLLM 2026 — Team SourcesŮres (French RAG Evaluation Challenge)

2026

Role: ML Engineer, four-person team; targeting both Task 1 (RAG QA) and Task 2 (attribution).

- Built a French-language RAG pipeline: document ingestion (PyMuPDF / pdfplumber), hybrid chunking, retrieval with FAISS, BAAI/bge-m3 embeddings, cross-encoder reranking, generation with Mistral-7B.
- Orchestration with LlamaIndex; reproducible evaluation harness with NDCG@10 and answer-attribution metrics.
- Team workflow: GitFlow branching (main / develop / feature/* / experiment/* / release/*), conventional commits, PR reviews.

Cost-Sensitive Credit Risk Prediction with Multilayer Bipartite Graphs — MSc Thesis

2023 – 2025

Best Scientific Poster Award, I3AFD 2026 Thematic School, University of Yaoundé I.

- Designed MNIFS (Multilayer Network Importance-based Feature Selection): automatic selection of predictive client descriptors from relational data.
- Modelled client behaviour with a bipartite multilayer graph and introduced a personalised PageRank variant to capture individual risk profiles.
- Reduced expected financial loss by over 70% across 7 benchmark datasets and 6 ML models under cost-sensitive evaluation.
- SHAP analysis: graph-derived features account for 66% of the top-10 most important variables.
- Extension submitted to *Data Mining and Knowledge Discovery (DMKD)*, February 2026.

Neural Networks from Scratch — NumPy

2026

- Implemented a full inheritance hierarchy of neural layers in pure NumPy: Dense, Dropout, BatchNorm, RNN, LSTM, GRU.
- Forward and backward passes, gradient checking, mini-batch training, and classic benchmarks for pedagogical and engineering clarity.

CONIA 2025 Hackathon — Plant Disease Classification (1st Prize)

2025

- CNN (YOLOv11s) reaching 98%+ accuracy with sub-50 ms inference.
- Deployed as an offline-first progressive web app (Nuxt 3, Vue 3, Vite, Flask).
- Publication: *International Journal of Computer Applications*, 186(78), 2025. DOI: 10.5120/ijca2025924642.

Multilingual Hate Speech Detection

2024

- Fine-tuned BERT / RoBERTa for French and English hate-speech classification; $F1 = 0.87$ on benchmark datasets.
- Served through a REST API for real-time inference.

Education

MSc, Data Science — University of Yaoundé I

2023 – 2025

GPA 3.17/4.0 (M2: 3.62/4.0, ranked 7th/28). Key courses: advanced data mining, computer vision, distributed learning, AI, algorithms and optimisation.

Professional Bachelor, Computer Science — University of Yaoundé I

2017 – 2020

GPA 3.16/4.0 (L3: 3.48/4.0, class valedictorian). Specialisation: information-systems security.

Awards & Distinctions

- **Best Scientific Poster**, I3AFD 2026 — cost-sensitive credit risk prediction with multilayer bipartite graphs. (2026)
- **Best Project**, I3AFD 2026 — French-language RAG pipeline. (2026)
- **1st Prize**, CONIA 2025 Hackathon — AI system for agriculture. (2025)
- **2nd Prize**, CANAM AI Competition — data analysis and ML. (2023)

Teaching & Community

- Academic tutor in data science and machine learning; supervised 5+ students on applied research. (2024 – 2025)
- Facilitator of the workshop “From data to KPIs: surfacing decision-grade insights”, 30+ participants, University of Yaoundé I. (2024)
- Local organising committee member, I3AFD 2026 Thematic School. (2026)

References available on request.